

ENTERPRISE CCTV MODERNIZATION

From ageing cameras to a 600+ camera security platform.

BALAJI WAFERS / MULTI-LOCATION VIDEO SURVEILLANCE



Designed, deployed and scaled by the D-TECH team

MAIN FACTORY / JAMNAGAR / HEAD OFFICE

A repeat-order success story built on resilient fiber, structured recording and controlled access.

EXECUTIVE SNAPSHOT

A security platform built to scale.

D-TECH transformed an unsatisfactory legacy camera environment into a structured multi-location CCTV architecture.

600+

CAMERAS

Across the broader deployed estate.

3

KEY LOCATIONS

Main Factory, Jamnagar and Head Office.

2

SERVER ROOMS

NVR racks installed for controlled operations.

6-CORE

OFC BACKBONE

CCTV today, capacity for future data needs.

DOCUMENTED ORDER MILESTONES

2019 / INITIAL MODERNIZATION

Purchase Order 4500225553 included 227 IP cameras, 10 NVRs, 80 surveillance hard disks, racks, PoE switching, CAT6 and a 6-core fiber backbone.

2021 / REPEAT EXPANSION

Purchase Order 4500319820 added 42 IP cameras, a 64-channel NVR, 8 surveillance hard disks, PoE switches, racks, CAT6 and additional 6-core fiber.

REPEAT ORDERS VALIDATED THE DESIGN

The supplied purchase orders document 269 cameras. Subsequent and additional orders expanded the overall installed environment beyond 600 cameras.

BUSINESS CHALLENGE

Old cameras were no longer meeting expectations.

Administration and security teams required better coverage, clearer ownership, scalable recording and a professional plant-wide design.

AGEING CAMERA ESTATE

Legacy cameras and inconsistent coverage were not delivering the visibility expected by plant administration and security.

MULTIPLE AREAS & USERS

The solution needed to cover plant, office and critical operational zones while serving different monitoring responsibilities.

SCALABILITY PRESSURE

Every new camera or area could not become a separate isolated network; the architecture needed room for repeated expansion.

RECORDING & RETENTION

Higher-resolution IP video increased demands on NVR channels, surveillance storage, bandwidth and rack infrastructure.

ACCESS GOVERNANCE

Plant administration and security required controlled viewing and administration rights based on responsibility.

FUTURE DATA NEEDS

The fiber investment needed to support CCTV immediately while protecting capacity for later business connectivity.

DESIGN QUESTION

How can one CCTV platform improve present security operations and remain ready for future locations, data services and repeat expansions?

D-TECH APPROACH

Survey first. Architect once. Scale repeatedly.

D-TECH treated CCTV as enterprise infrastructure - not simply a camera replacement exercise.

01**ASSESS**

Map blind spots, legacy limitations, monitoring expectations, cable routes, control rooms and risk areas.

02**DESIGN**

Create camera placement, bandwidth, PoE, rack, recording, storage and fiber architecture.

03**SEGMENT**

Separate surveillance traffic and define controlled access for administration and security users.

04**DEPLOY**

Install cameras, racks, NVRs, storage, switches, CAT6 and OFC with structured labeling.

05**VALIDATE**

Test views, night performance, recording, playback, fail paths and user permissions.

06**EXPAND**

Use the same standards and reserved fiber capacity for repeated orders and new locations.

D-TECH DELIVERY PRINCIPLE

Every camera, rack, switch, fiber path and access right is designed as part of one controlled surveillance environment.

SOLUTION ARCHITECTURE

A resilient plant-wide surveillance network.

Distributed camera and PoE infrastructure connects through a 6-core single-mode OFC backbone to controlled recording and monitoring zones.



MULTI-LOCATION STANDARD

The same design logic supports the Main Factory, Jamnagar and Head Office while allowing location-specific camera counts and access rights.

DEPLOYMENT EVIDENCE

Purpose-built recording infrastructure.

Actual site photographs show rack-mounted NVR and recording infrastructure installed for structured, maintainable operations.



NVR / RECORDING HARDWARE



SERVER-ROOM CCTV RACK

STRUCTURED INSTALLATION

Rack-mounted equipment, organized power and network connectivity, identification and maintainable service access.

DUAL SERVER-ROOM MODEL

Recording infrastructure distributed across two server rooms to support operational control, access governance and expansion.

ACCESS MANAGEMENT

Rights are assigned according to responsibility, ensuring appropriate live view, playback and administrative control for authorized users.

FUTURE-READY BACKBONE

One 6-core fiber. More than one purpose.

D-TECH designed the backbone to carry CCTV today while reserving capacity for future enterprise connectivity.

6-CORE SINGLE-MODE OFC BACKBONE

Three fiber pairs create a practical path for active surveillance and future services.

CORE 1 + 2**ACTIVE CCTV PATH**

Dedicated surveillance uplink for camera and NVR traffic.

CORE 3 + 4**FUTURE DATA / EXPANSION**

Reserved capacity for new areas, business data or alternate service paths.

CORE 5 + 6**RESILIENCE / GROWTH**

Available for redundancy, additional sites or future technology requirements.

SMART CAPITAL USE

A single planned fiber route avoids repeated civil work and protects the customer's infrastructure investment.

CLIENT CONFIDENCE

The forward-looking design demonstrated that D-TECH understood both immediate CCTV needs and future data growth.

REPEAT-ORDER JOURNEY

Confidence converted into expansion.

The supplied purchase orders show a clear progression from a major modernization program to additional camera and infrastructure orders.

MAR 2019

PO 4500225553

- 227 IP cameras
- 10 × 32-channel NVRs
- 80 × 6 TB surveillance drives
- 3,500 m of 6-core SM OFC

SEP 2021

PO 4500319820

- 42 additional IP cameras
- 1 × 64-channel NVR
- 8 × 8 TB surveillance drives
- Additional CAT6, racks, PoE and OFC

BROADER DEPLOYED ESTATE

600+

Expanded across repeated orders and multiple operating locations.

ORDER DOCUMENTS ARE SUPPORTING EVIDENCE

The case study summarizes the solution outcome; confidential pricing and full purchase-order copies are intentionally excluded.

BUSINESS IMPACT

A more capable and controllable security environment.

The value went beyond camera count: the project established a scalable security infrastructure standard.

BETTER VISIBILITY

Updated camera design improved coverage across operational, administrative and security-sensitive areas.

CENTRALIZED RECORDING

Structured NVR racks and surveillance storage created a more manageable recording environment.

CONTROLLED ACCESS

Administration and security users receive rights aligned with their roles and responsibilities.

SCALABLE EXPANSION

Standardized racks, switching, cabling and OFC simplify new camera additions and repeat orders.

FUTURE-READY BACKBONE

Reserved fiber cores protect capacity for future data services, alternate paths and new locations.

TRUSTED DELIVERY

Professional design, execution and support strengthened customer confidence in D-TECH as a long-term partner.

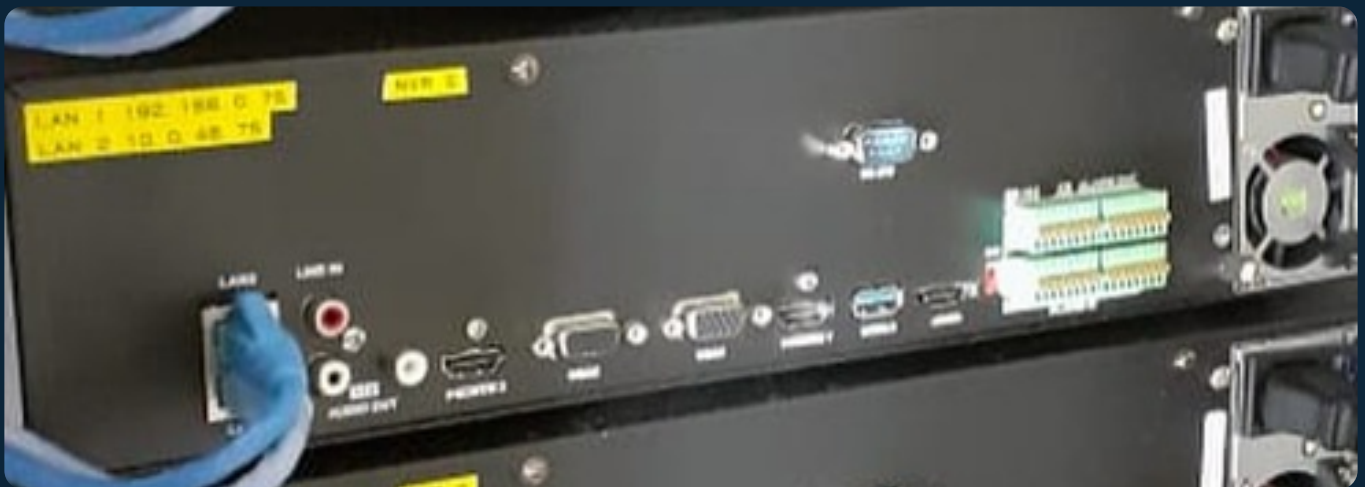
THE RESULT

A repeatable, multi-location CCTV architecture that scales from individual cameras to an enterprise-wide surveillance platform.

CASE STUDY CONCLUSION

Designed for today. Ready for tomorrow.

D-TECH converted a legacy CCTV challenge into a structured 600+ camera surveillance environment, supported by enterprise racks, NVR storage, a 6-core fibre backbone and controlled access across key locations.



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